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20.07.26

Experiment-01 Study of various Network commands

Aim

To study and practice various windows and Linux Network commands

Windows Commands

1) arp -a

Displays IP-to-MAC address mappings.

O/P

192.168.1.1

00-14-22-01-23-45

dynamic

2) hostname

Displays the Computer name

O/P

DESKTOP-ABC123

3) ipconfig/all

Displays complete Network configuration

O/P

IPV4 Address:

192.168.1.10 Subnet

Mask: 255.255.255.0

5) nbtstat -a <hostname>
Displays the NetBIOS name table of host
O/P
NetBIOS Remote
Machine Name Table

5) netstat
Displays active TCP connections
O/P
TCP 127.0.0.1:5000
127.0.0.1:51000
ESTABLISHED

6) netstat -r
Displays the IP routing Table
O/P
Network Destination
Netmask Gateway

7) nslookup google.com
Finds DNS/IP information for a domain
O/P
Name: google.com
Address: 142.250.X.X

8) pathping google.com
Shows the path to a destination and
packet loss information

o/p

0 PC -> 1 Gateway
-> ... -> google.com

9) ping google.com

Testes connectivity to the destination

o/p

Reply from
142.250.x.x : bytes = 32
time = ... TTL = ...

10) route print

Displays the windows routing table

o/p

IPV4 Route Table.

Linux commands

1) ip address show

Displays IP addresses and interfaces

o/p

inet 192.168.1.10/24

2) ifconfig

Displays network interface configuration

o/p

inet 192.168.1.10

netmask 255.255.255.0

3) ping google.com

Checks connectivity to a host

O/P

64 bytes from...

time = 20ms

4) mtr google.com

Shows route, delay and packet loss to a host

O/P

Loss% Snt Avg Best Wrst

5) tcpdump -D

Lists interfaces available for packet captures

O/P

1. eth0. 2. any 3. lo

6) tcpdump -i eth0 -c 10

Captures 10 packets from eth0.

O/P

10 packets captured

7) ip route show

Display IP routing table

O/P

default via

192.168.1.1 dev eth0

8) ip route get 8.8.8.8

Shows the route used to reach a particular IP

O/P

8.8.8.8 via
192.168.1.1 dev eth0

9) nmcli connection show

Displays available network connection profiles

O/P

Wired Connection 1
Ethernet eth0.

10) cat /etc/resolv.conf

Displays DNS server configuration

O/P

nameserver
192.168.1.1

Result

Thus various Windows and Linux networking commands were studied and executed.

28/4/26

28.07.26

Experiment - 02 Study of different types of Network Cables

Different type of Cables used in networking are

1. Unshielded Twisted Pair (UTP) Cable
2. Shielded Twisted Pair (STP) Cable
3. Coaxiale Cable
4. Fibre Optic Cable.

Cable Type	Category	Maxium Data Transmission	Advantages/ Disadvantages	Appeication Use
UTP	Category 3	10 bps	Advantages: → Cheaper in Cost → Easy to install as they have a smaller overall diameter Disadvantages: → More prone to EMI and noise	→ 10 Base-T Ethernet → Fast Ethernet Gigabit Ethernet
	Category 5	Upto 100 Mbps		
	Category 5e	1 Gbps		
STP	Category 6, 6a	10 Gbps	Advantages: → Shielded → Faster than UTP → Less susceptible to noise and interfaca Disadvantages: → Expensive → Greater installation effort	→ Gigabit Ethernet, 10G Ethernet (55m) → Widely used in data centres
SSTP	Category 7			

Coaxial Cable	RG-6 RG-59 RG-11	10-100 Mbps	Advantages: → High bandwidth → Immune to interference → Low loss bandwidth → Versatile Disadvantages: → Limited Distance → Cost → Size is bulky	→ Speed of Signal is 500m → Television network → High Speed internet Connections
Fibre optics Cable	Single mode Multi mode	100Gbps	Advantages: → High Speed → High bandwidth → High Security Disadvantages: → Expensive → Required Skilled installer	→ Maximum distance of Fibre optic Cable is around 100m

Result

Thus, the study of different types of Network cable is known.

11/8/24

4/8/26

Experiment-03 Cisco packet Tracer

Aim

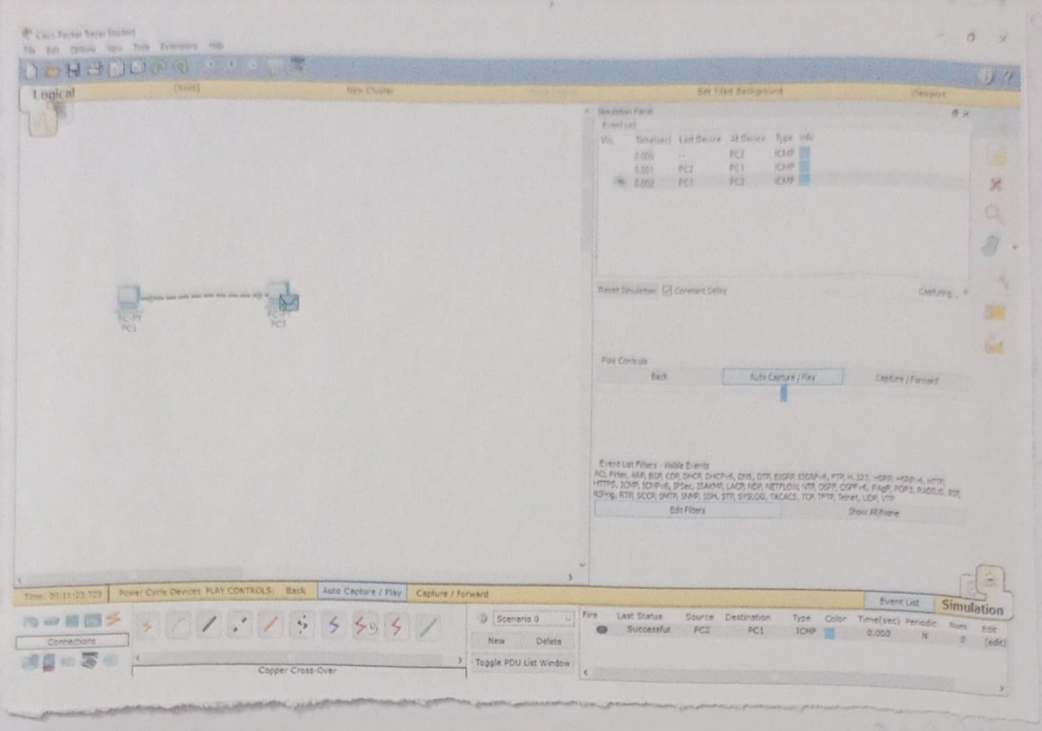
To study the cisco packet Tracer Simulation Tool, understand its user interface, design a simple network and analyze the behavior of hub, switch and peer to peer.

Cisco packet Tracer

Cisco packet Tracer is a network Simulation and modeling tool. It allow us to create and test computer networks without using real networking equipment. We can add PC's, switches, hubs, routers and other devices and observe how packets travel between them.

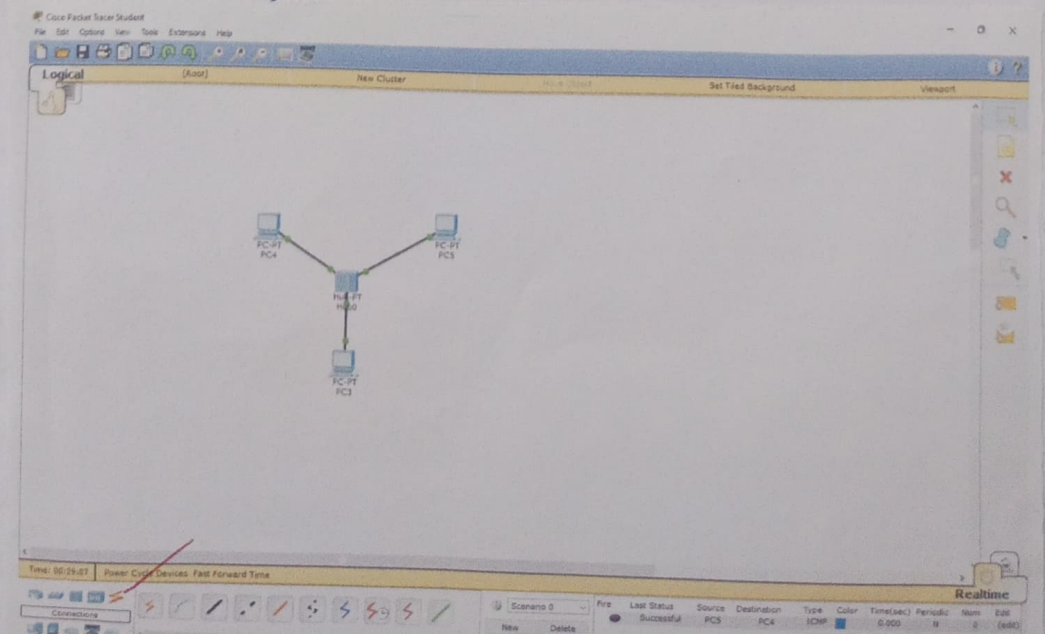
Peer to peer

A peer to peer network is a type of network in which all computers, called peers, have equal responsibilities and can communicate directly with each other without the need for a dedicated server. Each Computer can as both a client and a server by sharing resources such as files, printers and applications.



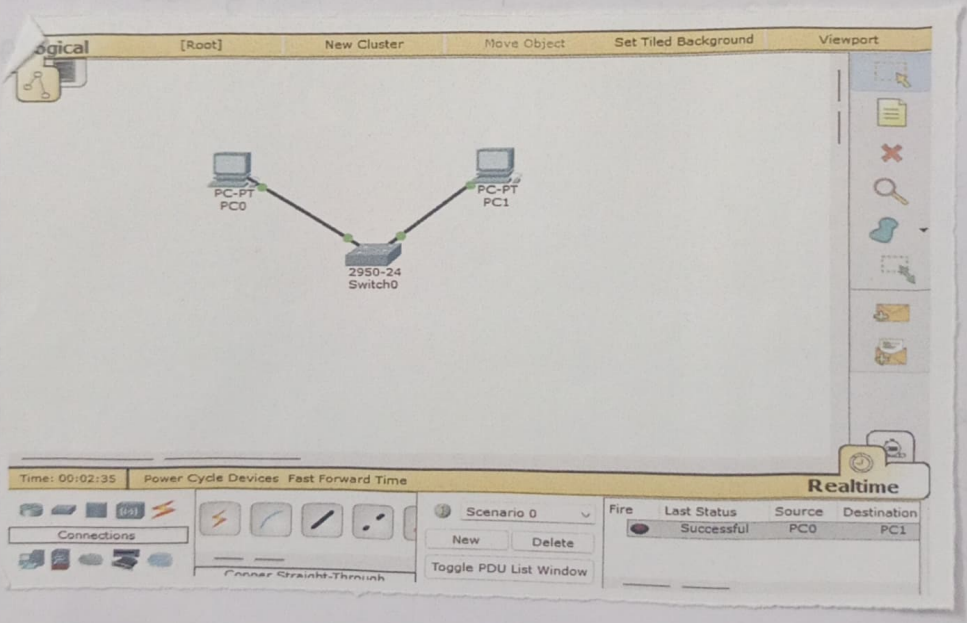
Hub

A Hub is a basic networking device used to connect multiple computers in a LAN. It operates at the physical Layer of the OSI model. When a hub receives data from one device, it broadcasts the data to all connected devices without checking the destination address. Only, the intended device accepts the data, while remaining devices ignore it.



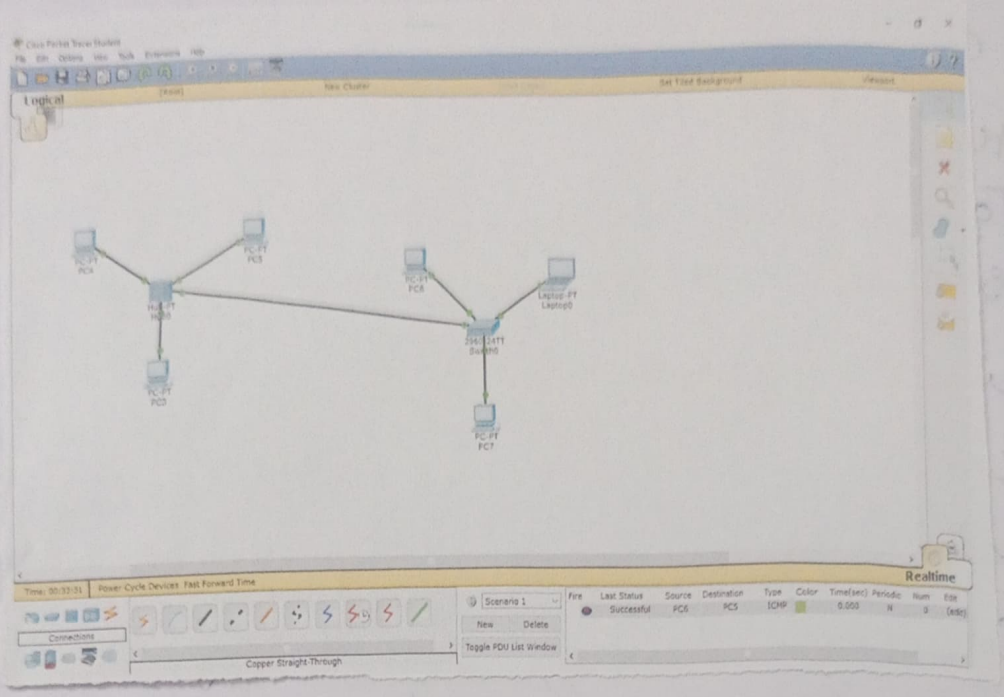
Switch

A switch is a networking device that connects multiple devices in a LAN and operates at the DLL of the OSI model. Unlike a hub, a switch forwards data only to the intended destination device by using the devices MAC address. It maintains a MAC address table to identify which device is connected to each port, resulting in faster communication, reduced network traffic.



Hub and Switch

Both hub and switch were used in the network packet forwarding behavior of both devices was observed and compared.



Result

Thus, peer to peer, hub & Switch networks were successfully designed and their communication behaviour were observed using cisco packet tracer

15/9/21

10/8/26

Experiment-04 Setup and configure a LAN using a switch and ethernet cables in your lab

Aim

To setup and configure a LAN using a switch and ethernet cables in your lab.

What is LAN?

LAN stands for Local Area Network which is a group of computers and connected devices linked together in a small physical location like a school, home or office building.

How to set up LAN?

Step 1: Connect the Hardware

-> Place the router in the Central location near a power outlet.

-> Connect your modem to the router's WAN or internet port using a ethernet cable

Step 2: Connect your devices

-> Wired setup: plug one end of ethernet cable into your computer network port and numbered LAN. ports on the router

Step 3 : Configure the router

→ Type your IP address 192.168.1.1 into the address bar and press Enter

→ Log In using the default username and password printed on the router

Step 4 : Check the connectivity

→ Check the connectivity between switch and other machine by using ping command in the command prompt of the device

Step 5 : Confirm the connectivity

Try to access the shared folder from other computers of networks

18/8/20

Experiment 05 : WireShark

Aim

To understand the features of wireshark, a packet capture tool and understand Encapsulation of information of various layers of a protocol.

Procedure

Step 1: Open wireshark and select the wifi network interface.

Step 2: Start the packet capture by double clicking the wifi interface.

Step 3: Observe the packets being captured in the wireshark window.

Step 4: Different packets such as ARP, DHCP, IGMPV3, NDNS and LLMNR can be observed in the packet list.

Step 5: Select any packet to view, its detailed information such as frame, Ethernet II, IP, TCP and Protocol details

Step 6: Apply the display filter in filter bar to display only DNS packet.

Step 7: Observe DNS query and respond's packet. The source and

8/26

Result

Thus, network packets was successfully captured and analyzed using wireshark and DNS packet were identified

7/9/26

Experiment 06 Error Detection and Correction using Hamming Code

Aim

Write a program to implement error detection and correction using HAMMING CODE concept. Make a test run to input data stream and verify error correction feature.

Code

Sender

```
data = input("Enter the data bits: ")
d1, d2, d3, d4 = map(int, data)
P1 = d1 ^ d2 ^ d4
P2 = d1 ^ d3 ^ d4
P4 = d2 ^ d3 ^ d4
code = [P1, P2, d1, P4, d2, d3, d4]
print("Hamming Code:", " ".join(map(str, code)))
```

Receiver

```
r = list(map(int, input("Enter received 7 bits: ")))
r1 = r[0] ^ r[2] ^ r[4] ^ r[6]
r2 = r[1] ^ r[2] ^ r[5] ^ r[6]
r3 = r[3] ^ r[4] ^ r[5] ^ r[6]
error = r1*1 + r2*2 + r4*4.
if error == 0:
    print("No error")
```

```
else:
    print("Error at position:", error)
    r[error-1] ^= 1
print("Corrected code:", ''.join(map(
        str, received)))
```

Input:

Enter 4 data bits : 1011

Sender Hamming code : 0110011

Received 7 bits : 0010011

Output:

Error at position : 2

Corrected code : 0110011

Result

Error detection and Error correction using Hamming code was executed successfully.

8/9/26

Experiment 07 Flow control using Sliding window protocol-

Aim

Write a program to implement flow control at data link layer using sliding window protocol.

Code (Sender)

```
import time
w = int (input ("window size :"))
msg = input ("Message :")
i = 0
while (i < len(msg)):
    end = min (1+w, len(Msg))
    with open ("Sender - Buffer", w)
                as f
        for j in range (i, end):
            f.write (f'{i} : {msg[j]}')
    Print ("Frame", ), "->", msg[j])
    time.sleep(2)
    with open ("RECEIVER - BUFFER")
                as f
        ack = f.read().
    i = int (ack[1])
Print ("Transmission Completed")
```

Receiver

```
import time
import os
expected 0
while true:
    if not os.path.exists ("Sender-Buffer"
        continue

    with open ("Sender_Buffer") as f
        frames = f.readlines()
    for x in frames:
        n = int (x.split(";")[0])
        if n == expected
            expected += 1
            print ("Received frame:",n
            with open("Received
                Buffer", "w") as f
                f.write ("ACK" + str
                    (expected))
        else:
            with open ("Receiver_Buffer
                "w" as f:
                f.write ("NACK" + str(expected)
            break
    time.sleep(1)
```

Result

The Sliding protocol was implemented successfully. ACK/NACK was simulated successfully.

22/9/26

Experiment 08 NMAP to discover live hosts using NMAP scans on the tryhackme platform

Aim

To discover live hosts using Nmap Scans (ARP, ICMP, TCP/UDP) on the Tryhackme platform

Procedure.

1. Setup and Environment establishment

Log in to TryHackMe platform and navigate to the target room: TryHackMe Nmap room.

2. ARP Scan for local host discovery

Perform an ARP request scan to identify active hosts on the local network segments.

Command:

Sudo nmap -PR -sn <target-IP subnet>

3. ICMP Echo ping Scan

Send ICMP echo Request packets to check for ICMP Echo reply responses from active hosts

Command:

Sudo nmap -PE -sn <target-IP | subnet>

4. ICMP timestamp ping Scan

Send TCMP timestamp request packets to discover hosts that blocks standard ICMP

5. ICMP address Mask Ping Scan,

Send ICMP address mask request packets to identify live signals by requesting subnet mask implementation

6. TCP ACK Ping Scan:

Send TCP ACK packets to click RST responses from active hat, bypassing firewall rules

Cmd:

Sudo nmap -PE -SN <target-IP/ subnet>

7. UDP Ping scan

Send empty UDP packets to destination port to detect path via ICMP path unreachable responses

Cmd:

<sudo nmap -PN -sn <target-IP/subnet>

9. Disable Port Scanning

Use the -sn flag with nmap to perform new discovery exclusively without triggering subsequent port scan

Cmd:

sudo nmap -sn <target-IP/subnet>

22/9/25

Result

Using TryHackme platform, use learned and executed to discover live hosts on NMAP scans (ARP, ICMP, TCP/UDP)

28/9/26

Experiment 09 Implementation Of Subnetting in cisco packet Tracer Simulator

Aim

To implement classless if subnetting using cisco packet Tracer and verify connectivity between different subnets

Procedure

1. Open cisco packet Tracer and Create a new network topology
2. Add routers, switches and PCs to the workspace
3. Connect the devices using appropriate Cables
4. Divide the network 192.168.1.0/24 into smaller subnets using 127 subnet mask
5. configure the IP address and subnets mask for the route interface.
6. Configure the PC's with suitable IP address, subnet masks and default gateways.

⑦ Configure the Switch ports and router interface

⑧ Use the PING command to test connectivity between the devices

⑨ Verify that the packets are successfully transmitted between the connected networks

Network	Subnet Mask	Router IP	PC IP Range
192.168.1.0 127	255.255.255.224	192.168.1.1	192.168.1.11. 15
192.168.2.0 127	255.255.255.224	192.168.2.1	192.168.2.11. 15
192.168.3.0 127	255.255.255.224	192.168.3.1	192.168.3.11. 15
192.168.4.0 127	255.255.255.224	192.168.4.1	192.168.3.11. 15

28/9/22

Result

The classless IP subnetting was successfully implemented using CISCO packet Tracer.

Exp 12 a) End to End communication at Transport layer.
29/9/26

Aim

To implement an ECHO SERVER application using TCP Sockets.

Code

```
import socket
s = socket.socket()
s.bind (("localhost", 5000),
s.listen(1)
c, adder = s.accept()
msg = c.recv(1024).decode
Print ("Client:", msg)
c.sent(msg.encode)
c.close()
s.close()
```

Echo client

```
import socket
s = socket.socket()
s.connect(("localhost", 5000))
msg = input ("Enter Message:")
s.send(msg.encode())
Print ("Echo:", s.recv(1024).decode())
s.close()
```

Input

Enter message : Hello Server

Output

Server

Client connected

Received : Hello Server

Client

Echo from Server : Hello Server

29/9/22

Result

The ECHO client server using TCP Socket was studied and executed Successfully.

b) Implement chat client Server using TCP/IP socket

Aim

To implement a chat client Server application using TCP sockets

Code

```
import Socket
S = socket.socket()
s.bind (("Local host, 5000))
S.listen()
C, a = s.accept()
while True:
    msg = c.recv(1024).decode()
    print ("Client:", msg)
    if msg == 'bye':
        break
    reply = input ("Server")
    C.send(reply.encode())
C.close()
```

Client

```
import socket
s = socket.socket()
s.connect(("localhost",5000))
while True:
    msg = input("Client :")
    s.send(msg.encode())
    if msg == "bye":
        break
    print("Server:",s.recv(1024).
                    decode())
s.close()
```

Input

Client : Hello
Server : Hi
Client : Bye

Output

Client : Hello
Server : Hi

29/9/22

Result

The chat client server application were successfully studied and

Exp 13 Implement your own
30/9/26 ping program

Aim

To implement a ping program using python to check the connectivity between source and destination host

Code

```
import os
import time
host = input ("Enter host:")
for i in range(4):
    start = time.time()
    result = os.system("Ping -n 1 "
        + host + " >null")
    end = time.Time()
    if result == 0:
        print ("Reply from", host,
        "Time=", round ((end-
                start) *
            1000, 2 : "ms"))
    else:
        print ("Request timed out")
    time.sleep()
```

Input:

Enter host : 127.0.0.1

Output

Reply from 127.0.0.1 Time = 1.23 ms

Reply from 127.0.0.1 Time = 0.98 ms

30/9/23

Result :

The ping program was successfully studied and implemented.

Exp 14 Packet Sniffing using Raw Sockets
01/00/26

Aim

To write a program using raw sockets to capture and display network packet.

Code.

```
import Socket
s = Socket.Sock(Socket.AF_INET,
    Socket.SockRaw, Socket.
    IPPROTO_IP)
s.bind(("0.0.0.0",0))
s.setSockOpt(Socket.IPPROTO_IP,
    Socket.IP_HDRNCL, 1)
Print("Packet Sniffer Started...")
while True:
    data.add = s.recv from(65535)
    print("Packet received from:",
        addr[0])
```

Input

Network packets

Output:

```
Packet Sniffer started...
Packet received from: 192.168.1.5
Packet received from: 192.168.1.1
Packet received from: 142.120.183.14
```

01/10/24

Result

The RAW SOCKET SNIFFING PROGRAM was successfully implemented and network packets was captured and displayed.